PAPAGARI SUMANTH REDDY

AI Engineer | Generative AI, RAG & Agentic Systems | Backend (Python, FastAPI)

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PROFESSIONAL SUMMARY

AI engineer with an M.Tech from **IIT Hyderabad** (GATE CS/IT top 5%) who shipped a **production LLM document-automation agent** and cut **MILP solver runtime by up to 71.3%** at Greenko Group. Builds RAG and agentic systems end to end: hybrid retrieval, evaluation, FastAPI microservices, self-hosted LLM serving, Docker and AWS deployment.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash, TypeScript
Generative AI: RAG, Agentic AI, LangChain, LangGraph, Model Context Protocol (MCP), Hybrid Search (BM25 + dense, RRF), Reranking, HyDE, Corrective RAG, Self-RAG, Text-to-SQL
LLMs & ML: vLLM (Qwen), AWS Bedrock (Claude), OpenAI API, Ollama, Gemma, PyTorch, Hugging Face Transformers, BERT, CLIP
Backend & Data: FastAPI, REST APIs, Microservices, PostgreSQL, MySQL, MongoDB, Redis, Keycloak (OAuth2/OIDC, RBAC), React
Vector Databases: Qdrant, Pinecone, pgvector, sqlite-vec, Weaviate
Cloud & DevOps: AWS (Lambda, App Runner, Elastic Beanstalk, ECR, EKS, S3), Docker, Kubernetes, GitHub Actions CI/CD, Linux, Git
Evaluation & Optimization: RAGAS, LangSmith, Opik, pytest, Gurobi (MILP), Ray

WORK EXPERIENCE

Greenko Group | AI & Software Engineering Intern, Gen AI / Operations Research — Jan 2026 – Jun 2026

- Shipped the **GIMS AI Agent to production** for the IMS documentation unit: a 5-pipeline **FastAPI** platform that restructures ISO/IMS procedures and generates native, editable Word (DrawingML) swimlane flowcharts, replacing manual rebuilds and outsourcing.
- Built an **agentic chat assistant** on self-hosted **vLLM (Qwen)** with web search for standards lookup, plus document restructuring on **AWS Bedrock (Claude)** and low-latency retrieval on **PostgreSQL pgvector**, streamed to users over SSE.
- Delivered a Dockerized **React + TypeScript** multi-session interface behind nginx with **Keycloak OIDC SSO**, role-based access control, per-user allowlisting, file validation and fail-fast configuration checks.
- Cut Day-Ahead Market **MILP solver runtime by up to 71.3%** (40–50% on average, 8 benchmark cases) by warm-starting **Gurobi** with prior-day and same-weekday solutions across 96 time blocks, using **Ray** and MinIO persistence.
- Root-caused silent warm-start failures to contract-driven constraint changes and built a validate-first safeguard with automatic cold-start fallback, backed by automated benchmark tests.

PROJECTS

KathaMitra: Offline AI Storytelling Mentor (M.Tech Thesis) | *Gemma 2B, MediaPipe, sqlite-vec, Android* — May 2025 – May 2026

- Built a fully on-device Android mentor for 9–12-year-olds using a quantized **Gemma 2B** LLM, local vector retrieval and on-device STT/TTS; **100% local inference**, DPDP Act 2023 compliant, scored **4.80/5.0** in pedagogical evaluation.
- Designed a Symmetric Context retrieval step for code-switched Hinglish speech and tuned the pipeline to **under 450 ms** end-to-end latency (60% reduction).

Multi-Source RAG + Text-to-SQL Copilot | *FastAPI, Pinecone, PostgreSQL, Redis, AWS Lambda* | GitHub — 2026

- Built an 18-endpoint service that routes each question to **document RAG, Text-to-SQL (Vanna) or a hybrid path**, with SQL safety validation and an approval workflow before execution.
- Added two-level caching (**Redis** query cache, SHA-256 document cache on S3) designed to cut repeat-query LLM cost by 40–60%; deployed on **AWS Lambda (ARM64)** via Docker, ECR and **GitHub Actions**, with Opik tracing and RAGAS evaluation.

Corrective + Self-Reflective RAG | *FastAPI, Qdrant, OpenAI, Tavily, Docling, AWS Elastic Beanstalk* | GitHub — 2026

- Implemented **hybrid search** in Qdrant (dense + BM25 sparse vectors fused with RRF) with HyDE and reranking; added **CRAG** relevance grading with web-search fallback and a **self-reflective loop** that refines queries until answers are grounded in sources.

Multimodal RAG Document Intelligence Pipeline | *GLM-OCR, GPT-4o, Qdrant, FastAPI, Ollama* | GitHub — 2026

- Built a PDF-to-answer pipeline: layout detection across 23 element types, OCR for tables and formulas, **GPT-4o figure captioning**, hybrid Qdrant retrieval and pluggable rerankers (BGE, Jina, Qwen VL), runnable in the cloud or fully local via Ollama.

Production RAG Q&A Service | *FastAPI, LangChain, Qdrant, LangSmith, RAGAS, AWS App Runner* | GitHub — 2026

- Shipped an async, streaming document Q&A API with source attribution, **RAGAS** faithfulness and relevancy scoring, LangSmith tracing, **70%+ test coverage** and a GitHub Actions pipeline deploying Docker images through ECR to AWS App Runner.

MCP-Powered Code Review & Bug Fix Assistant | *LangChain, MCP, Docker* | GitHub — 2026

- Developed an agentic code reviewer that uses **Model Context Protocol** tools to read repositories, detect bugs and propose fixes.

Scientific Diagram Captioning | *PyTorch, CLIP, ViT, BART* — Aug 2024 – Nov 2024

- Trained a multimodal captioning system on a 4M-image SciCap subset, reaching **73% Recall@10** on image-text alignment.

EDUCATION

Indian Institute of Technology Hyderabad | M.Tech, Techno-Entrepreneurship | CGPA 8.73/10 — Jul 2024 – May 2026

Malla Reddy College of Engineering | B.Tech, Computer Science and Engineering | CGPA 7.50/10 — Jul 2020 – May 2024

ACHIEVEMENTS & LEADERSHIP

- **GATE 2024 (CS/IT):** ranked in the top 5% of 123,000+ candidates nationwide.
- **Japan collaborations (2025):** startup and SDG 11 projects with student founders at Nagoya, Kyoto and Shimane Universities.
- **Hospitality Coordinator**, SPIC MACAY 10th International Convention, IIT Hyderabad (2024 – 2025).